AITS
ALL INDIA TEST SERIES

Dropper JEE Advanced

AJPA11.2/10 **Test- 10**

DURATION : 180 Minutes **DATE : 16/03/2025** **M.MARKS : 198**

Topics covered

Physics:	Full Syllabus
Chemistry:	Full Syllabus
Mathematics:	Full Syllabus

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration.
3. The test booklet consists of 54 questions. The maximum marks are **198**.

SECTION 1 (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from **0 TO 9, BOTH INCLUSIVE.**
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

  Full Marks : +3 If ONLY the correct integer is entered:
  Zero Marks : 0 If the question is unanswered;
  Negative Marks : –1 In all other cases.

SECTION 2 (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:

  Full Marks : +4 If only (all) the correct option(s) is(are) chosen;
  Partial Marks : +3 If all the four options are correct but ONLY three options are chosen:
  Partial Marks : +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;
  Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
  Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
  Negative Marks : –2 In all other cases.

SECTION 3 (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

  Full Marks : +4 If ONLY the correct numerical value is entered;
  Zero Marks : 0 In all other cases.

OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. Do not fold or make any stray mark on the Answer Sheet (OMR).

Name of the Students (In CAPITALS) : _____

Roll Number : _____

OMR Bar Code Number : _____

Candidate's Signature : _____ Invigilator's Signature _____

IMPORTANT CONSTANTS

Speed of light in free space,	:	$3.00 \times 10^{8}\ ms^{-1}$
Permeability of free space,	:	$4\pi \times 10^{-7}\ Hm^{-1}$
Permittivity of free space,	:	$8.85 \times 10^{-12}\ Fm^{-1}$
The Planck constant,	:	$6.63 \times 10^{-34}\ Js$
Rest mass of electron,	:	$9.1 \times 10^{-31}\ kg$
Rest mass of proton,	:	$1.67 \times 10^{-27}\ kg$
Molar gas constant,	:	$8.31\ JK^{-1}\ mol^{-1}$
The Avogadro constant,	:	$6.02 \times 10^{23}\ mol^{-1}$
The Boltzmann constant,	:	$1.38 \times 10^{-23}\ JK^{-1}$
Gravitational constant,	:	$6.67 \times 10^{-11}\ N\ m^2kg^{-2}$
Acceleration of free fall	:	$9.8\ ms^{-2}$
Rydberg Constant	:	$1.097 \times 10^{7}\ m^{-1}$
Atomic mass unit	:	$1.67 \times 10^{-27}\ kg$
Charge on proton	:	$1.6 \times 10^{-19}\ C$

IMPORTANT VALUES

$\sqrt{2}$ = 1.414

$\sqrt{3}$ = 1.732

$\sqrt{5}$ = 2.236

π = 3.142

e (Euler's constant) = 2.718

ln 10 = 2.303

$\log_{10}2$ = 0.3010

$\log_{10}3$= 0.4770

$\log_{10}7$ = 0.845

**** Use above values unless otherwise specified in a question.***

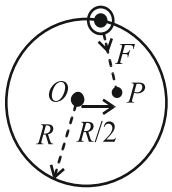
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PART-I (PHYSICS)

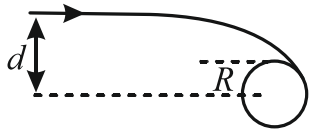
SECTION-1

Single Digit Integer Type Questions:

1. In free space, a small bead of mass m can move on a smooth circular fixed wire (radius R) under the action of a conservative force $F = \frac{Km}{r^2}$ directed (r is position of bead from P and K is a constant) towards a point P within the circle at a distance $R/2$ from the center. What should be the minimum velocity (in m/s) of bead at the point of the wire nearest to the center of force (P) so that bead will complete the circle. (Take $\frac{K}{3R}$ = 8) (Assume all quantities are in SI units)

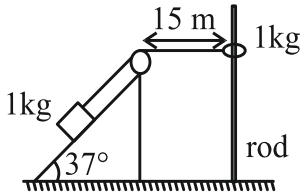


2. A comet moving with a velocity $\frac{1}{2}\sqrt{\frac{GM}{R}}$ in the interstellar space starts approaching a planet of mass M and radius R from infinity and passes while grazing the surface of the planet. Neglecting the effect of gravity of other heavenly bodies, the ratio of aiming parameter and radius of planet, $\frac{d}{R}$ is

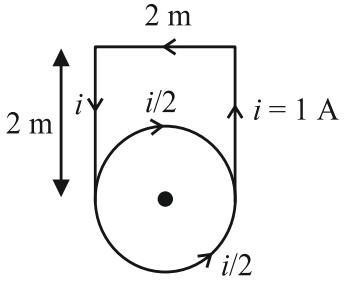


3. A small particle of mass m moves in such a way that the potential energy $U = \frac{1}{2}mb^2r^2$, where b is a constant and r is the distance of the particle from the origin (Nucleus). Assume Bohr model of quantization of angular momentum and circular orbits is applicable. If the radius of nth allowed orbit is proportional to $n^{4/k}$, then value of k is :-

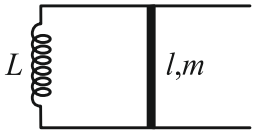
4. A light string attached to a bead passing over a smooth fixed pulley at a distance of d = 15 m from the vertical rod as shown in the figure. At the other end of the string a block of mass 1 kg is attached, lying over a smooth fixed inclined plane of inclination angle 37°. Mass of the bead is 1 kg and friction between bead and vertical rod is zero. The bead is held in level with the pulley and released. If the speed of the bead when the string between bead and pulley makes an angle 53° with the horizontal is $\sqrt{\frac{\alpha \times 1000}{41}}$ m/s, find value of α. (Take g = 10 m/s^2)



5. Find the value of magnetic dipole moment (in A m^2) of the following circuit the current distribution is shown in the diagram



6. Two parallel resistance less rails are connected by an inductor of inductance L at one end as shown in figure. A magnetic field B exists in the space which is perpendicular to the plane of the rails. Now a conductor of length l and mass m placed transverse on the rail and given an impulse j towards the rightward direction. Velocity of the conductor is half of the initial velocity after a displacement of the conductor $d = \sqrt{\frac{xj^2L}{4B^2l^2m}}$. Find the value of x



SECTION-2

One or More Than One Correct Type Questions:

7. A non-uniform thin rod *AB* of mass *m* and length *l* can be pivoted to a smooth hinge passing through any point on the rod so that it can freely rotate in the vertical plane when released.

CASE I : The rod is hinged at end *A* and it is released from rest with rod making an initial angle of 53° with the vertical ,we noticed that initial angular acceleration of the rod is $\frac{g}{l}$.

CASE II : The rod is hinged at end *B* and it is released from rest with rod making the same initial angle of 53° with the vertical, but now we noticed that initial angular acceleration of the rod is $\frac{2g}{l}$. (Where *g* is magnitude of accelerations due to gravity). Which of the following are correct?

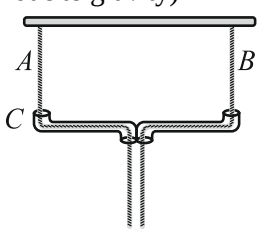
(A) Centre of mass of the rod is a distance of $\frac{l}{4}$ from end B.

(B) Moment of inertia of the rod about an axis perpendicular to the rod passing through centre of mass is $\frac{3}{80}ml^2$

(C) Magnitude of initial acceleration of the centre of mass of the rod in CASE I is $\frac{3g}{4}$

(D) Magnitude of initial acceleration of the centre of mass of the rod in CASE II is $\frac{g}{2}$

8. Two identical ropes *A* and *B* of uniform mass per unit length are suspended from the ceiling and are made to pass through a metallic tubular structure *C* as shown in the figure. The structure is made by welding two identical bent tubes. There is no friction between the ropes and the tubes and the structure is almost inertia-less. Initially the structure is held at rest and then released. What will happen? (*g* = acceleration due to gravity)



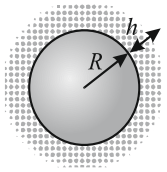
(A) It will not move at all

(B) It will move downwards with speed decreasing at the rate 0.5*g*.

(C) It will move downwards with speed increasing at the rate 0.5*g*.

(D) The rope will heat up.

9. During fair weather conditions atmosphere of the Earth can be considered as a good conductor with resistivity ρ depending on the height *h* from the surface of the Earth as $\frac{1}{\rho} = a + bh^2$, where *a* and *b* are constants. It is also known that near the surface of the earth electric field is E_0 directed towards the center of the earth which causes a radial current in the atmosphere. Assume E_0 is constant in magnitude and $h << R$.



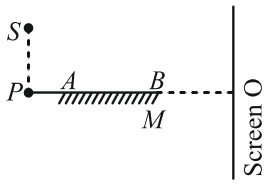
(A) The current density *J* decreases with increase in height.

(B) The electrons will be travelling in radially outward direction.

(C) The charge density will be negative at any general height.

(D) The charge density at height *h* is $\frac{2\varepsilon_0 E_0}{R+h}$ in magnitude.

10. In a Lloyd's mirror experiment a narrow slit *S* is transmitting a light of wavelength λ is placed 3 mm above a small plane mirror (as shown). The light coming directly from the slit and that coming after the reflection interfere at a screen placed at a distance of 90 mm from the slit. Length of mirror is 2 mm and the middle point of mirror is 2 mm from point *P*. If fringe width is 0.1 mm, then



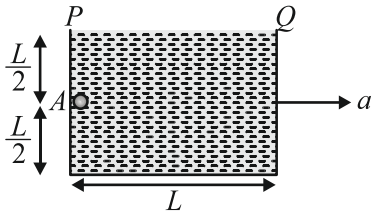
(A) the wavelength of light used in the experiment is $\frac{10}{3}$ μm

(B) the number of fringes obtained on the screen is 1800

(C) the number of fringes obtained on the screen is 1500

(D) the wavelength of light used in the experiment is $\frac{20}{3}$ μm

11. A small solid ball of density ρ is held inside at point *A* of a cubical container of side *L*, filled with an ideal liquid of density 4ρ as shown. The container is closed at the top. Now, if the container starts moving with constant acceleration *a* horizontally and the ball is released from point *A*, simultaneously, then: (g = acceleration due to gravity)



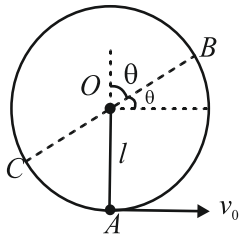
(A) For the ball to hit top of container at end *Q*, $a = 3g$

(B) For the ball to hit top of container at end *Q*, $a = 2g$

(C) Ball hits the top of container at end *Q* after a time $t = \sqrt{\frac{L}{3g}}$

(D) Ball hits the top of container at end *Q* after a time $t = \sqrt{\frac{L}{2g}}$

12. A heavy particle is attached to one end of a light string of length *l* whose other end is fixed at *O*. The particle is projected horizontally with a velocity v_0 from its lowest position *A*. When the angular displacement of the string is more than 90°, the particle leaves the circular path at *B* located at angle θ from horizontal. The string again becomes taut at *C* such that *B, O, C* are collinear.



(A) $\theta = 30°$

(B) $\theta = 45°$

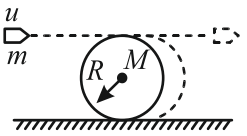
(C) $v_0 = \sqrt{\left(\frac{4+3\sqrt{2}}{2}\right)gl}$

(D) $v_0 = \sqrt{\left(\frac{6-2\sqrt{3}}{3}\right)gl}$

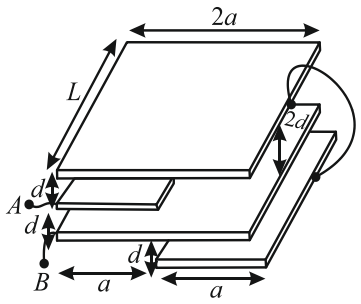
SECTION-3

Numerical Value Type Questions:

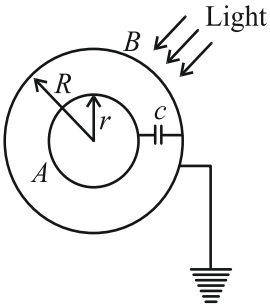
13. A bullet of mass *m* moving with velocity *u* just grazes the top of a sufficiently rough solid cylinder of mass *M* and radius *R* resting on a rough horizontal surface as shown in the figure. Assume that the cylinder rolls without slipping. The angular velocity of the cylinder is $\omega = \frac{4mu}{(8m+aM)R}$ and the final velocity of the bullet is $v = \frac{8mu}{(8m+bM)}$, then the value of $(a + b)$ is _____.



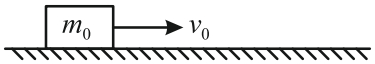
14. Four thin metallic plates have rectangular shapes with dimensions $L \times a$ or $L \times 2a$. All those plates are placed parallel to each other with equal spacing *d* between consecutive neighbouring plate as shown at the picture below. Top and bottom plates are connected with conducting wire. Determine effective capacitance C_{AB} measured between contacts *A* and *B* connected to the middle plates. If $C_{AB} = \frac{x\varepsilon_0 aL}{d}$, find *x*.



15. A conducting solid sphere of radius $r = 2$ cm is surrounding by a conducting shell which is made up of fine wire mesh with radius $R = 4$ cm as shown in fig. The shell and sphere are joined by a capacitor $C = 2\ \mu F$. Work function of material of sphere *A* and shell *B* is 1 eV and 4 eV respectively. Light falling on this system have frequency 7.27×10^{14} Hz. The energy stored in capacitor *C* in steady state condition (in micro joule) is $\left(\frac{h}{e} \times 7.27 = 30 \times 10^{-15}\ \text{J/A}\right)$ (neglect capacitance of sphere itself)

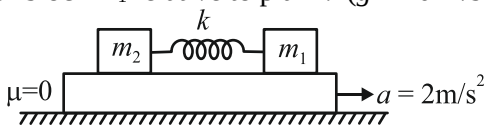


16. At $t = 0$ An ice cube of mass m_0 is given a velocity v_0 on a rough horizontal surface with coefficient of friction μ. The block is at its melting point and latent heat of fusion of ice is L. The block receives heat only due to the friction forces and all work is converted into heat. The mass of the remaining ice block after time t is $m = m_0 e^{-\frac{a\mu g}{L}\left(v_0 t - b\mu g t^2\right)}$. Find the value of $(a + b)$.



17. If the velocity of light (c), gravitational constant (G) and Planck's constant (h) are chosen as fundamental units, then if the dimensional formula of mass in this new system is $\left[c^x G^y h^z\right]$. Find the value of $4(x + y + z)$

18. Two blocks of masses m_1 = 1 kg and m_2 = 2 kg are connected by a spring of constant k = 100 N/m and placed over a plank as shown. The coefficient of friction between m_2 and plank is 0.5 and there is no friction between m_1 and plank. Initially the whole system, is at rest. The plank is now pulled to right with constant acceleration a = 2 m/s^2. Find the amplitude (in cm) of oscillation of block m_1 relative to plank. (g = 10 m/s^2)



PART-II (CHEMISTRY)

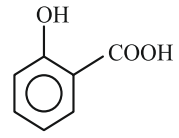
SECTION-1

Single Digit Integer Type Questions:

19. In an aqueous solution, mole fraction of solute is 0.2 and its density is d_1(g/ml). It is diluted such that mole fraction of solute becomes 0.1 and density becomes d_2 (g/ml). If density of new solution is 20% less than initial solution then molarity of new solution is almost half of the original solution. If molar mass of the solute is 'X' gm/mol, then closest integer value of $\left[\frac{X}{13}\right]$ is

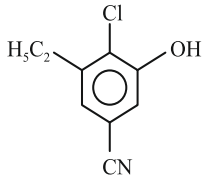
20. The no. of compounds with correct IUPAC name is/are _____.

(a)



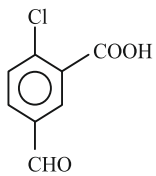
2-Carboxyphenol

(b)



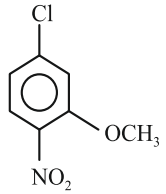
3-Ethyl-4-chloro-
5-hydroxybenzenecarbonitrile

(c)



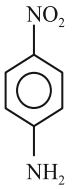
3-Formyl-
5-chlorobenzenecarboxylic acid

(d)



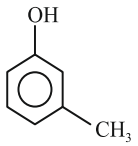
1-chloro-3-methoxy-
4-nitrobenzene

(e)



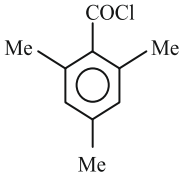
4-Amino-
1-nitrobenzene

(f)



3-Methylphenol

(g)



2,4,6-Trimethylbenzene-
carbonylchloride

21. Find the number of molecules which are planar

(a) ClF_3 (b) SF_4 (c) XeF_6 (d) XeF_2
(e) XeF_4 (f) H_2O (g) H_2S (h) NH_3
(i) PH_3 (j) PCl_5 (k) PCl_3 (l) OCl_2

22. Find number of substances which produce H_2 gas when react with cold water.
Ca, CaH_2, Mg, $MgCl_2$, Ba, BaO_2, $NaHCO_3$, Ca_3N_2.

23. $H-(CH_2)_n-\overset{\overset{O}{\|}}{C}-OH \xrightarrow[HVZ]{Cl_2/P_4}$ Product/s

Total number of products (including stereoisomers, if any) formed in above reactions for n = 0, 1, 2 are

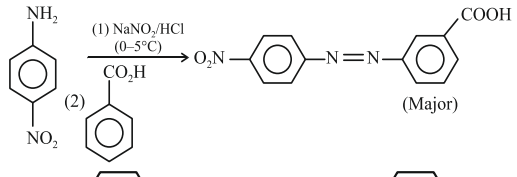
24. The number of correct statements among the following is _____.
(i) Bleaching action of SO_2 is through reduction
(ii) Bleaching action of Cl_2 is through oxidation
(iii) Iodine is more soluble in ethanol as compared to water
(iv) Iodine is more soluble in water containing KI, but solubility decreases on continuous addition of I_2
(v) Alkali metals are at the ascending part of Lother Meyer's Curve.
(vi) Alkaline earth metals are at the descending part of Lother Meyer's Curve.
(vii) Halogens are at the peak of Lother Meyer's Curve.

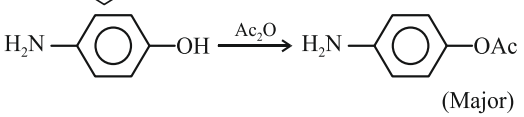
SECTION-2

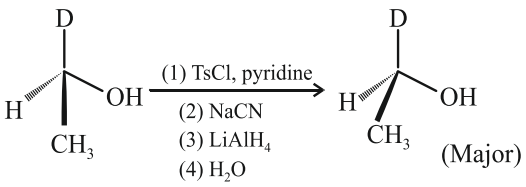
One or More Than One Correct Type Questions:

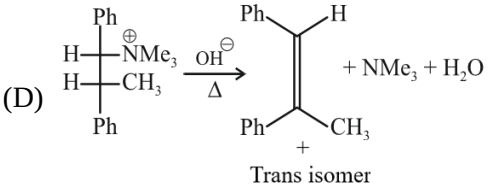
25. Which of the following statements are correct?
(A) Ionic carbide are formed by highly electropositive metals
(B) Carborundum and boron carbide are covalent compounds
(C) Carborundum and boron carbide are used as abrasive
(D) Al_4C_3 and CaC_2 on hydrolysis gives C_2H_2

26. Among the following reactions, identify the INCORRECT ones.

(A) 

(B) 

(C) 

(D) 

27. Consider the complex
P: $[ML_6]^{x-}$ has crystal field stabilising energy $= -1.2\Delta_0$
P is paramagnetic in nature on the basis of CFT. Select the correct combination(s) of metal (M) and (L) [L is monodentate ligand, x may be +ve or –ve integer] (Neglect pairing energy if any)

	M	**L**
(A)	Cr^{3+}	H_2O
(B)	Fe^{3+}	CN^-
(C)	Ni^{2+}	NH_3
(D)	Ni^{2+}	H_2O

28. Which of the following statement(s) is/are correct?
(A) In the NaCl structure, each Na^+ ion is surrounded by 6 Cl^- ions as nearest neighbours and twelve Na^+ ions as next nearest neighbours.
(B) The co-ordination number of cation generally increases as the radius ratio increases.
(C) Packing efficiencies higher than 0.74 are possible with cation and anion of different sizes.
(D) During a crystal defect, electrical neutrality should be maintained.

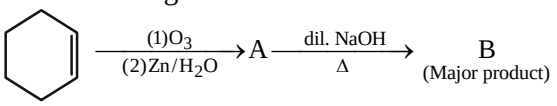
29. Hydrated ferric chloride ($FeCl_3.6H_2O$) cannot be converted to anhydrous $FeCl_3$ on heating but forms a compound (P). When $FeCl_3.6H_2O$ is treated with (Q) then anhydrous $FeCl_3$ is formed along with SO_2. Identify compounds (P) and (Q).
(A) (P) = FeOCl
(B) (Q) = $SOCl_2$
(C) (P) = Fe_2O_3
(D) (Q) = SO_2Cl_2

30. Which of the following statement(s) is/are correct about polymers?
(A) Neoprene is a polymer of isoprene.
(B) Glyptal is a condensation polymer of ethylene glycol and phthalic acid.
(C) Terylene is a condensation polymer of ethylene glycol and terephthalic acid.
(D) Nylon–6,6 is a condensation polymer of hexamethylenediamine and adipic acid.

SECTION-3

Numerical Value Type Questions:

31. In the following reaction



$$\xrightarrow[(2)Zn/H_2O]{(1)O_3} A \xrightarrow[\Delta]{\text{dil. NaOH}} \underset{\text{(Major product)}}{B}$$

The molecular weight (in g mol^{-1}) of the major product (B) is x. The value of $\frac{x}{5}$ is [Nearest integer]

32. When formaldehyde is treated with ammonia, a crystalline compound urotropine is formed. If the number of carbon atoms and nitrogen atoms in the compound are x and y respectively, then the value of $\frac{x}{y}$ is

33. A gas of mass 1 kg undergoes a quasi-static expansion, which follows P = a + bV where a and b are constants. P is in kPa and V is in m^3/Kg. The initial and final pressure are 10^3 kPa and 200 kPa and corresponding volumes are 0.2 m^3 and 1.2m^3. Work done by the gas during process is –x kJ. Calculate the value of $\frac{x}{100}$.

34. The molecular formula of Glauber's salt is $Na_2SO_4.10H_2O$. How much mass (in grams) of oxygen is present in 32.2 g of the hydrated salt? (Given, molar mass of Na-23, S -32, O-16, H-1 g/mol)

35. Under strongly acidic conditions, one mole nitrobenzene on electrolytic reduction gives a product (x) and consumes 'n' faraday of charge while under mild acidic conditions, one mole of nitrobenzene, on electrolytic reduction gives product (y) and consumed 'm' faraday of charge. So, the value of $\frac{m}{n}$ is _____.

36. The weight of CH_4 (in grams) produced by reacting 4.6 grams of glycerol with excess of CH_3MgBr is (Molar mass of C-12, O-16, H-1 g/mol)

PART-III (MATHEMATICS)

SECTION-1

Single Digit Integer Type Questions:

37. The value of $\lim_{n\to\infty}\left({}^{2n}C_n\right)^{1/n}$ is _____

38. The number of points where $f(x) = ||\ln|x|| - 2|$ is non-derivable is (in $x \in (-\infty, \infty)$)

39. Number of real solutions for the equation

$$\begin{vmatrix} \sin^{-1}x & \cos^{-1}x & \tan^{-1}x \\ \cos^{-1}x & \tan^{-1}x & \sin^{-1}x \\ \tan^{-1}x & \sin^{-1}x & \cos^{-1}x \end{vmatrix} = 0 \text{ equals}$$

40. Let $H_n = 1+\frac{1}{2}+\frac{1}{3}+....+\frac{1}{n}$ and

$T_n = \frac{1}{(n+1)H_nH_{n+1}}$. If $T_1 + T_2 + T_3 + ... + T_{n-1}$

$= \frac{\sum_{r=a}^{n}\frac{1}{r}}{\sum_{s=b}^{n}\frac{1}{s}}$, where $a, b \in N$ then $a - b$ equals

41. If α, β are the roots of the equation $x^2 - 2x + 3 = 0$, then the constant term of the equation (with leading coefficient unity) whose roots are $\alpha^3 - 3\alpha^2 + 5\alpha - 2$ & $\beta^3 - \beta^2 + \beta + 5$ is_____.

42. If $\overrightarrow{f(x)} = (\cos x)\hat{i} + (\sin x)\hat{j} + (\cos 2x)\hat{k}$, $\overrightarrow{g(x)} = (\tan x)\hat{i} + (\sin 3x)\hat{j} + (\cos 4x)\hat{k}$ and $\overrightarrow{h(x)} = (\cos 3x)\hat{i} + (\sin 5x)\hat{j} + (\cos 6x)\hat{k}$ are three variable vectors given, then $\int_0^{\pi}\left[\overrightarrow{f(x)}\ \overrightarrow{g(x)}\ \overrightarrow{h(x)}\right]dx$ where [.] denotes scalar triple product, is equal to

SECTION-2

One or More Than One Correct Type Questions:

43. Let $n \geq 1$ be an integer and $I_n = \int_0^{\pi}\frac{1-\cos nx}{1-\cos x}dx$, then

(A) I_1, I_2, I_3 ... is an A.P.
(B) I_1, I_2, I_3.... is a G.P.
(C) $I_1 = \pi$
(D) $I_1 = 0$

44. If the value of

${}^{2000}C_2 + {}^{2000}C_5 + {}^{2000}C_8 + {}^{2000}C_{11}.... + {}^{2000}C_{2000}$

is $\frac{2^{k_1}-1}{k_2}$, then

(A) $k_1 - 666k_2 = 1$ (B) $k_1 - 666k_2 = 2$

(C) $k_1 - 13^{k_2} = 197$ (D) $k_1 - 13^{k_2} = -197$

45. Let a line L passing through a point (–1, –2, 3) is perpendicular to two lines.
$\frac{x-1}{2}=\frac{y-1}{1}=\frac{z-1}{4}$ and $\frac{x-0}{1}=\frac{y-2}{3}=\frac{z-1}{2}$.
If perpendicular distance of a point on line L. from the plane $x+y+z-3=0$ is $2\sqrt{3}$, then the point(s) is/are
(A) (–3, –2, 4) (B) (–5, –2, 5)
(C) (17, –2, –6) (D) (–7, –2, 6)

46. A curve $y = f(x)$ whose differential equation is given by $x\,dy + 4\ln x\,dx = 2(y+1)dx$ passes through (1, 3), then
(A) $\left[f\left(\sqrt{2}\right)\right]$ is equal to 6 (where [.] is G.I.F)
(B) $\left[f\left(\sqrt{3}\right)\right]$ is equal to 8 (where [.] is G.I.F)
(C) function $y = f(x)$ is increasing for $x \in (0, \infty)$
(D) function have exactly one point of inflection in $(0, \infty)$

47. If the area of region bounded by tangents at ends of latus rectum of ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ and directrices of hyperbola $\frac{x^2}{9}-\frac{y^2}{72}=1$ is A, then which of the following statement(s) is/are correct.
(A) If $A=a-b\sqrt{5}$, then $ab = 8$
(B) If $A=a-b\sqrt{5}$, then $ab = 9$
(C) $[A] = 9$, (where [.] is G.I.F)
(D) $[A] = 10$, (where [.] is G.I.F)

48. Let $M = [a_{ij}]_{3\times2}$ and $N = [b_{ij}]_{2\times3}$ be two matrices such that $(MN)^2 = 3MN$ and $\det(NM) \neq 0$, then choose the correct option(s) (P is 2 × 2 matrix)
(A) Det$(NM) = 9$
(B) If $P(NM) = I$, then
$\lim\limits_{n\to\infty} \det(P + P^2 + P^3 \ldots\ldots P^n) = \frac{1}{8}$
(C) If $P(NM) = I$, then
$\lim\limits_{n\to\infty} \det(P + P^2 + P^3 \ldots\ldots P^n) = \frac{1}{4}$
(D) Det $(NM) = 3$

SECTION-3

Numerical Value Type Questions:

49. Let $n(A) = 4$ then number of relations $A \to A$ which are reflexive is 2^{λ}. The number of divisor of λ is

50. Sum of the series $\cot^{-1}\left(2\sqrt{2}\right)+\cot^{-1}\left(4\sqrt{2}\right)+\cot^{-1}\left(7\sqrt{2}\right)+\cot^{-1}\left(11\sqrt{2}\right)+\ldots.$ upto infinite terms is $\tan^{-1}(k)$, then $\frac{k^2}{4}$ is equal to

51. The mean of some even number of observations is 17 if alternate observations are increased by 2 then new mean is

52. The solution set of inequality $(2 – 3\sin x – 2\sin^2 x) < 0$ in the interval $(0, 2\pi)$ is $(a\pi, b\pi)$ then $a + b$ is

53. The area of region by two branches of $(y-x)^2 = x^3$ and straight line $x = 1$ is

54. If the function $f(x)=-4e^{\frac{1-x}{2}}+1+x+\frac{x^2}{2}+\frac{x^3}{3}$ and $g(x) = f^{-1}(x)$ then value of $g'\left(\frac{-7}{6}\right)$ equals

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